SEMICONDUCTOR

INTRODUCTION

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Atomic Structure

Conductor, Insulator, and Semiconductor

Properties of Semiconductor

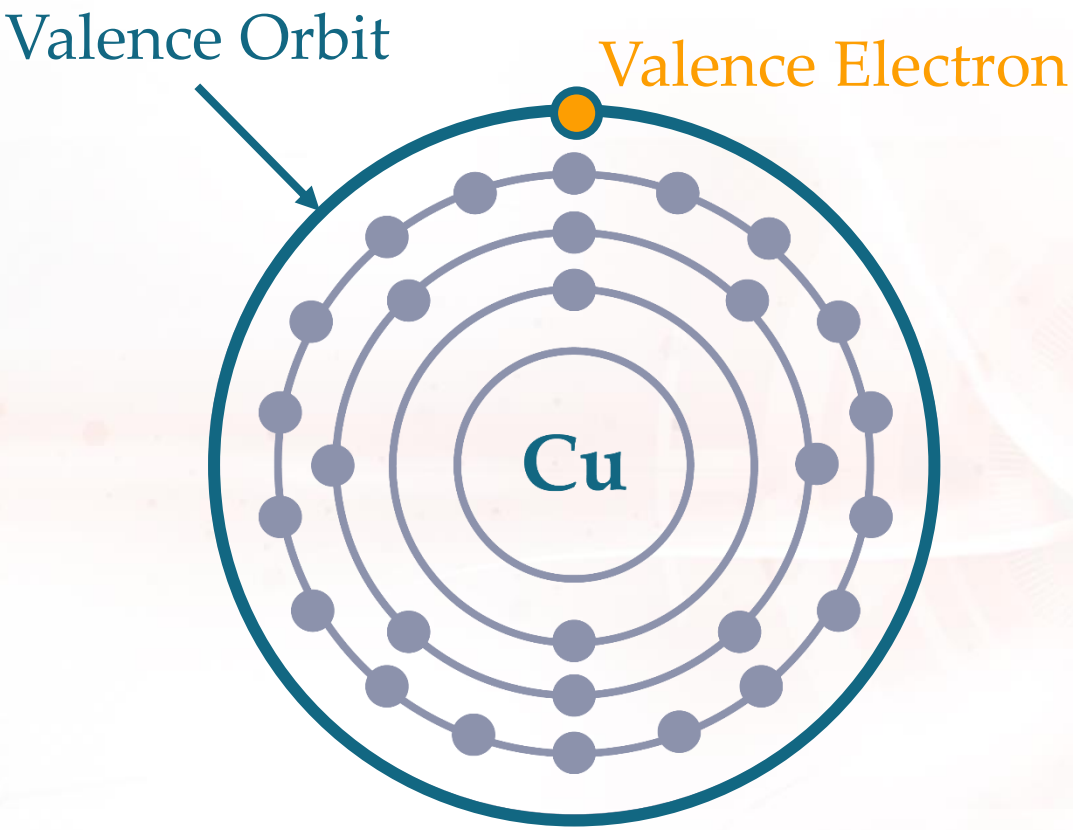
Intrinsic and Extrinsic

Two Types of Extrinsic Semiconductor

N-Type and P-Type

ATOMIC STRUCTURE

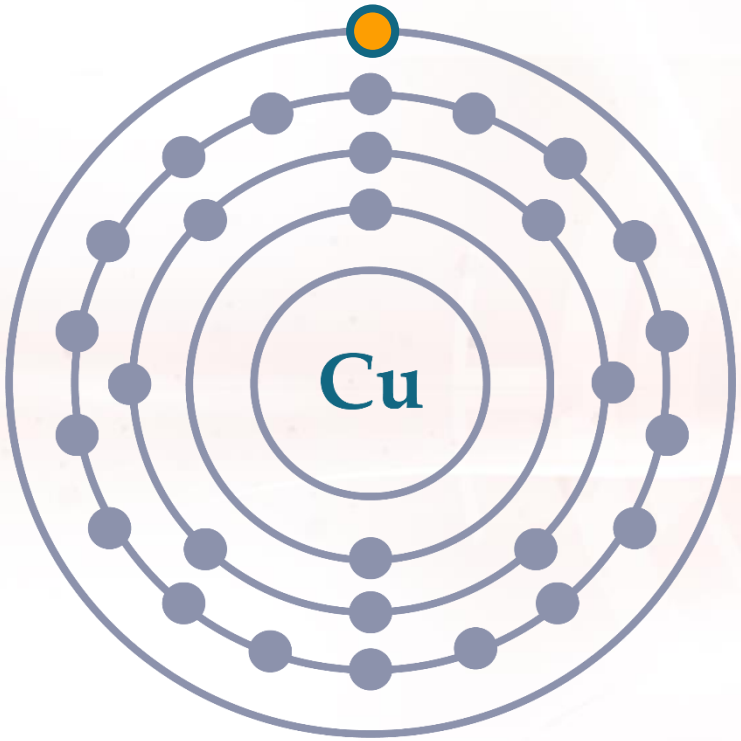
VALENCE ORBIT



The **outermost** electron shell or energy level of an atom

Controls the **electrical properties** of the atom

CONDUCTOR



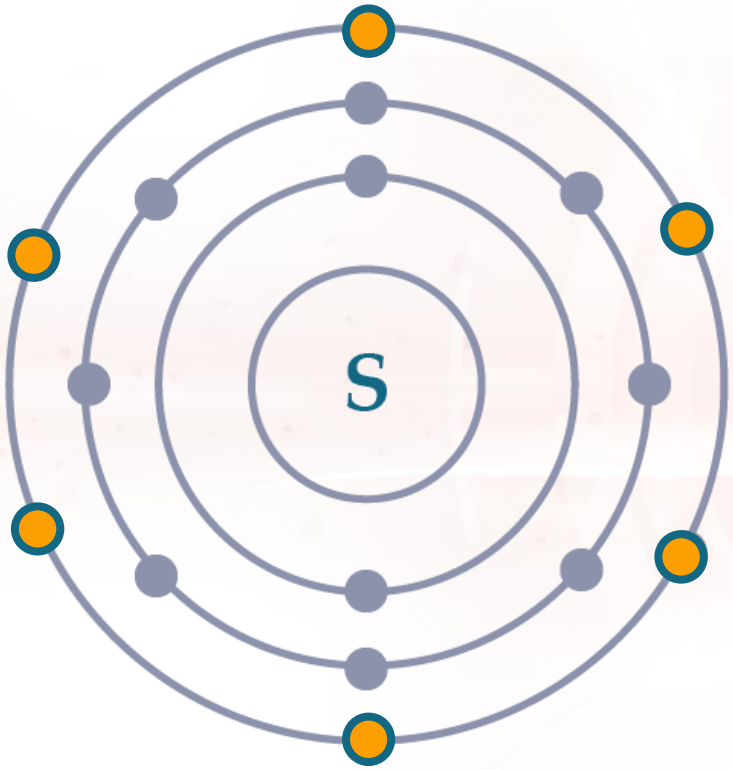
Materials that **allow** the electric current to flow through them

- **Copper**
- Silver
- Gold

Valence Electrons

- **Less** than 4 electrons

INSULATOR



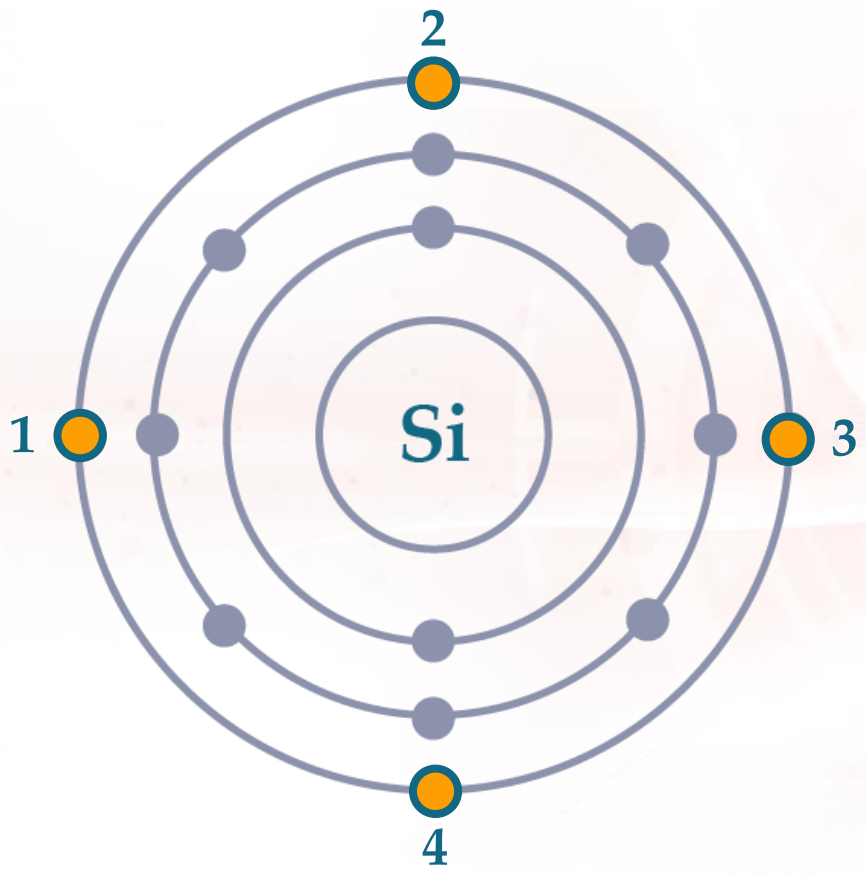
Materials that **do not** conduct electricity

- **Rubber**
- Plastic
- Sulfur

Valence Electrons

- **More** than 4 electrons

SEMICONDUCTOR



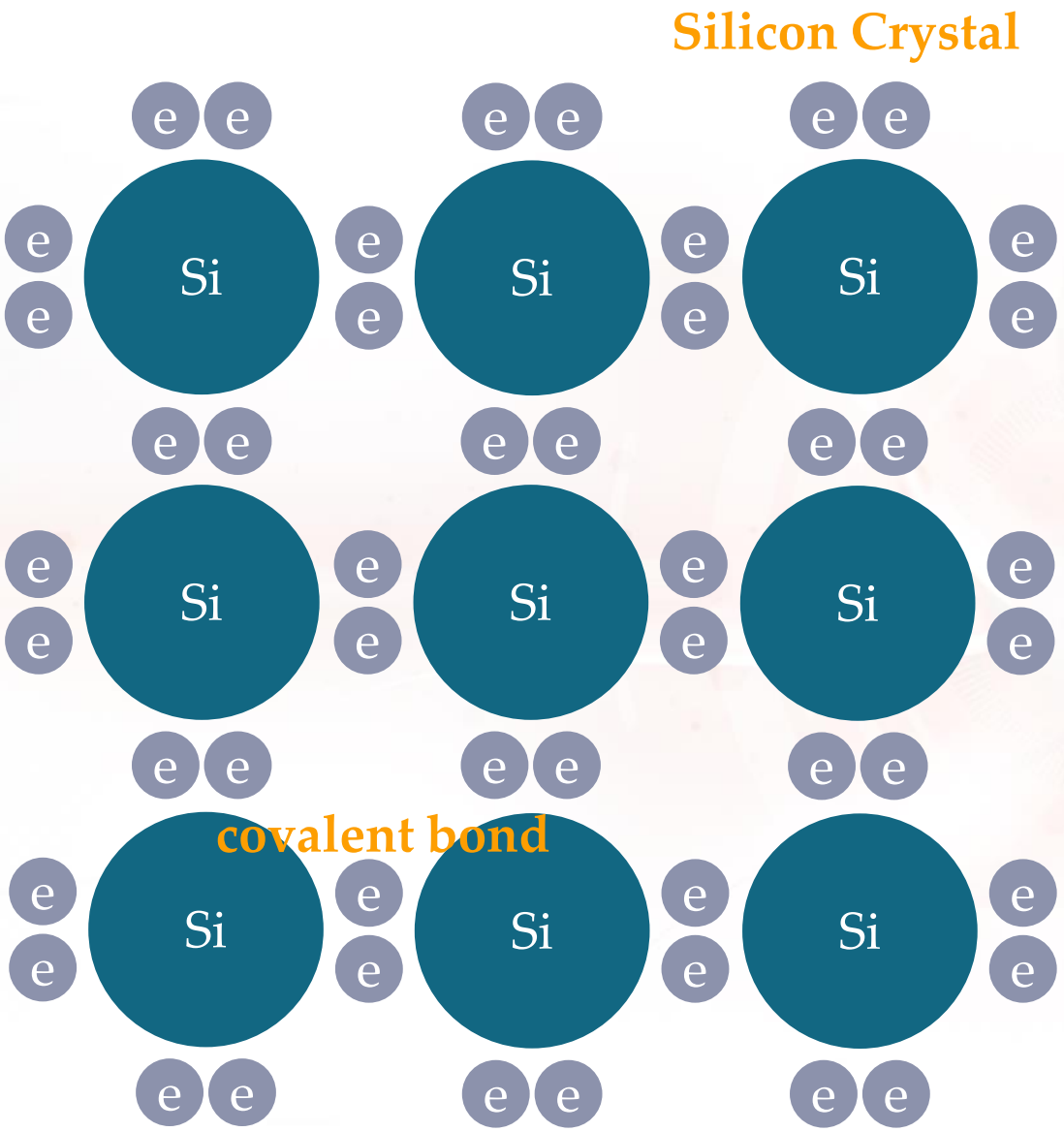
Electrical properties **between** conductor and insulator

- **Silicon**
- Carbon
- Germanium

Valence Electrons

- **Exactly 4 electrons**

PROPERTIES OF SEMICONDUCTOR



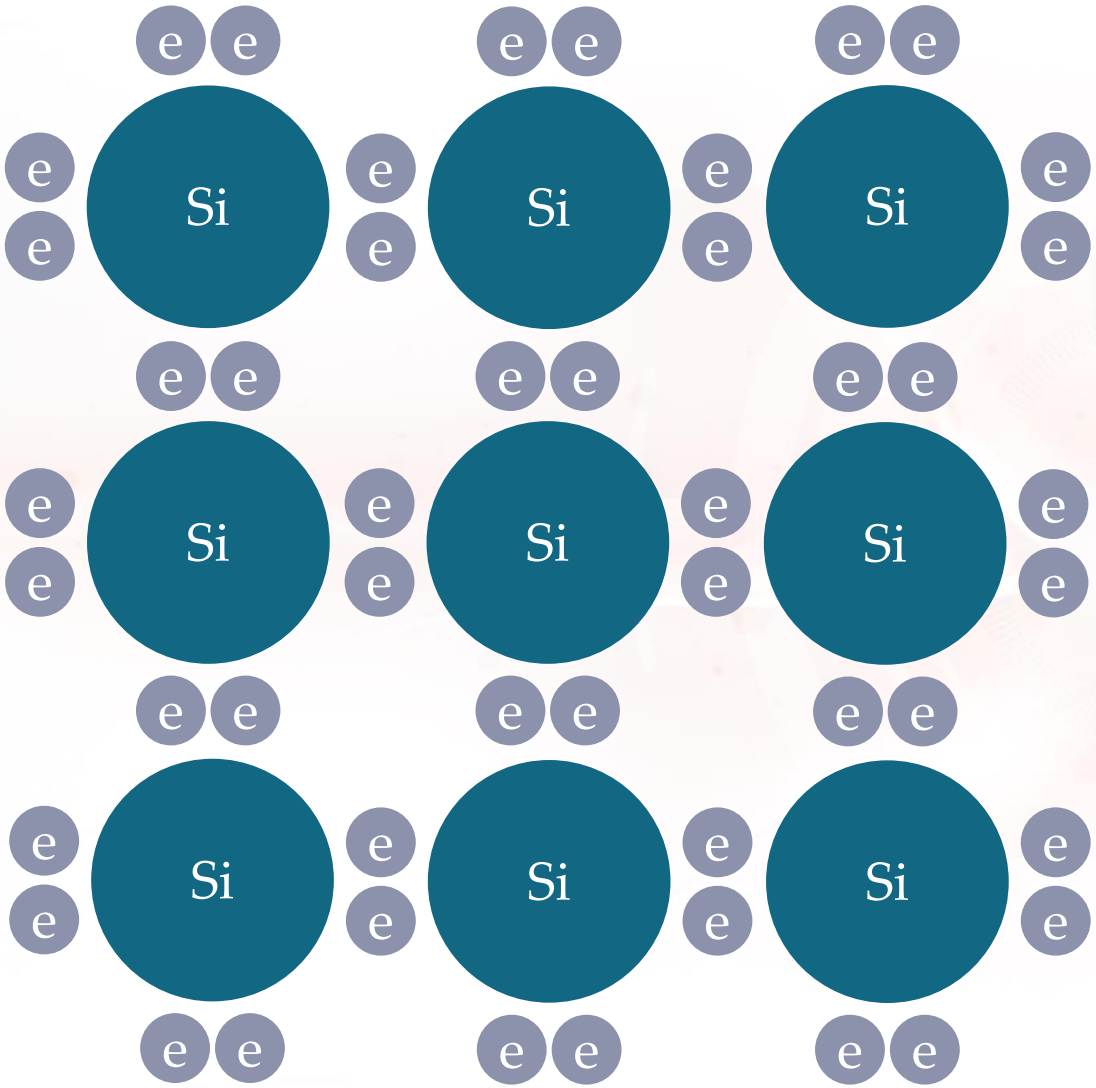
Valence Saturation

Valence electron = 8

Chemical Stability

The valence orbit can hold no more than 8 electrons

INTRINSIC SEMICONDUCTOR



Pure Semiconductor

Acts like an **insulator** at room temperature

EXTRINSIC SEMICONDUCTOR

Doping

Adding **impurity** atoms to an intrinsic crystal to alter its electrical conductivity

Group III Elements

B Al Ga In

Group V Elements

P As Sb

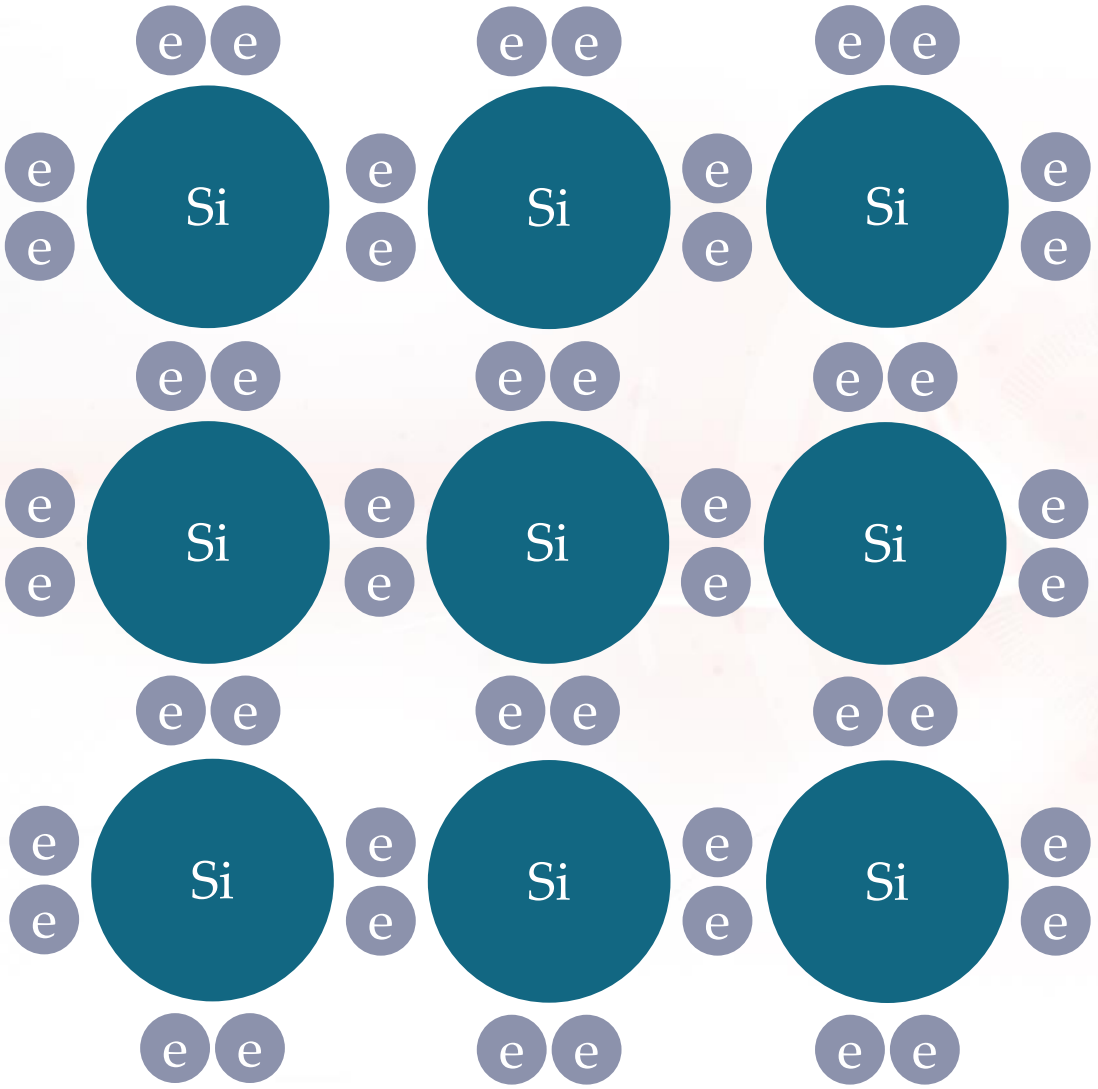
TWO TYPES OF EXTRINSIC SEMICONDUCTOR

TWO TYPES OF FLOW

Electron Flow

Hole Flow

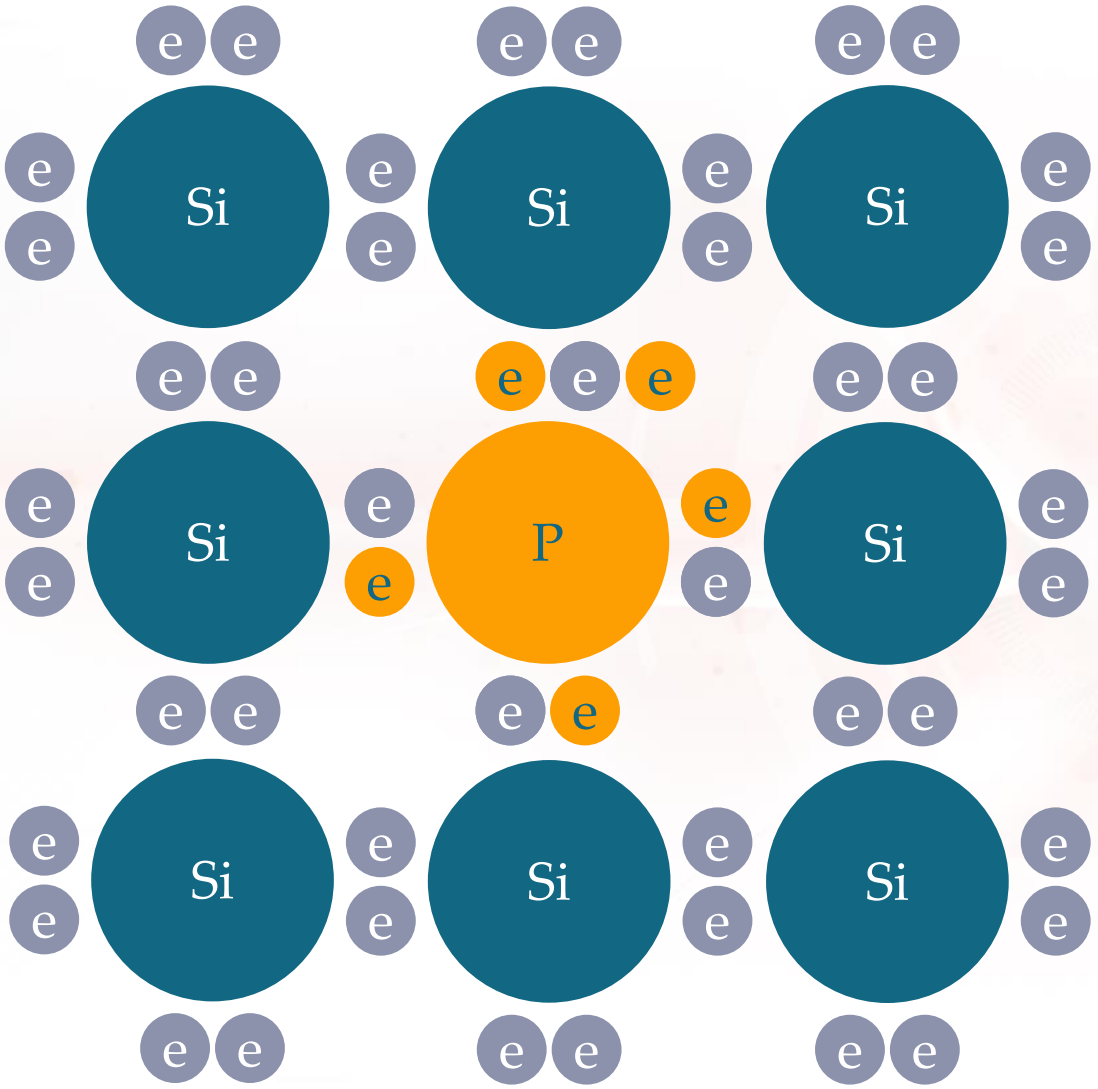
N-TYPE SEMICONDUCTOR



Doped with **pentavalent** impurity

- Arsenic (As)
- Antimony (Sb)
- Phosphorus (P)

N-TYPE SEMICONDUCTOR

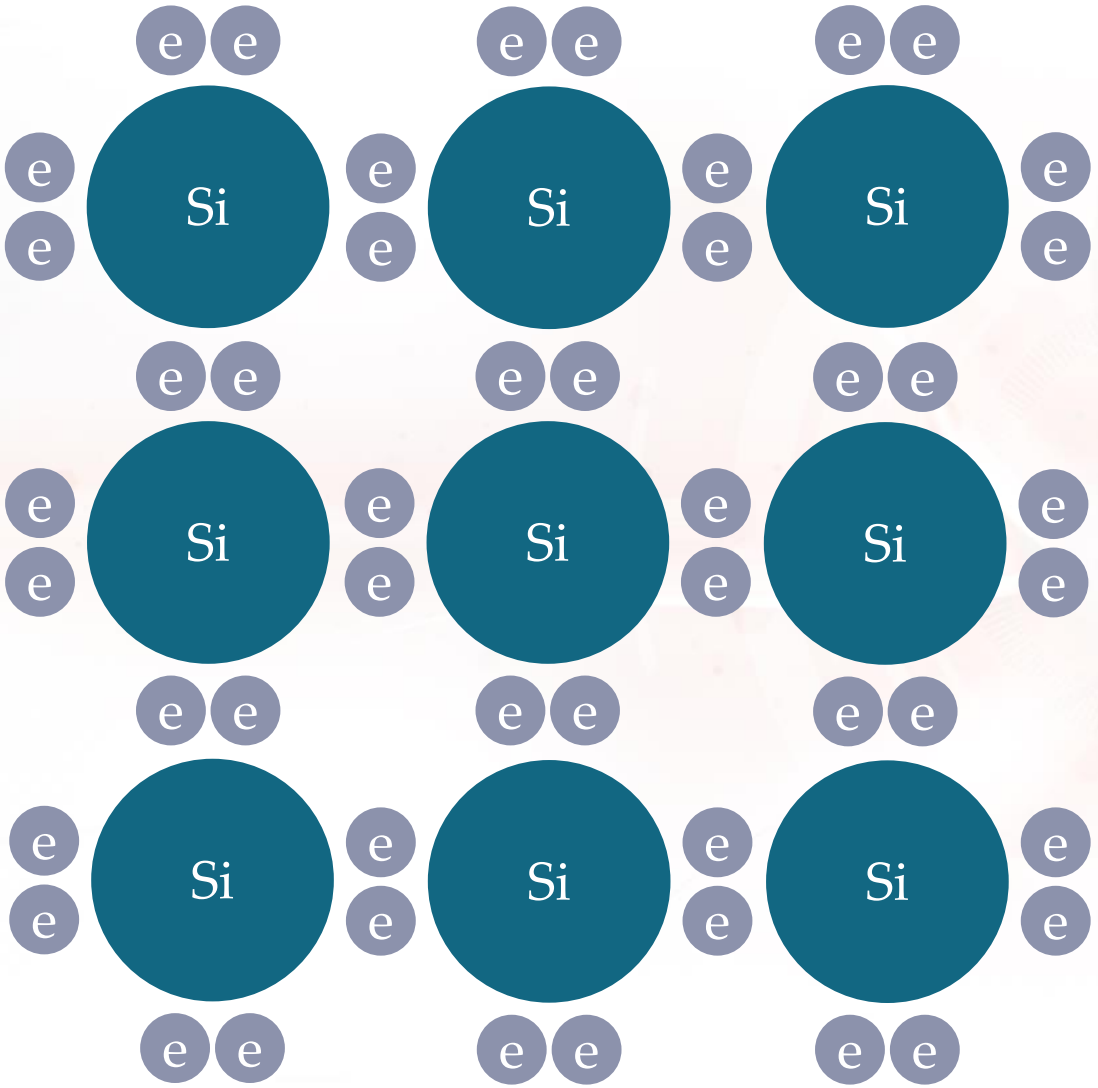


Doped with **pentavalent** impurity

- Arsenic (As)
- Antimony (Sb)
- Phosphorus (P)

Majority Carriers	Minority Carriers
electrons	holes

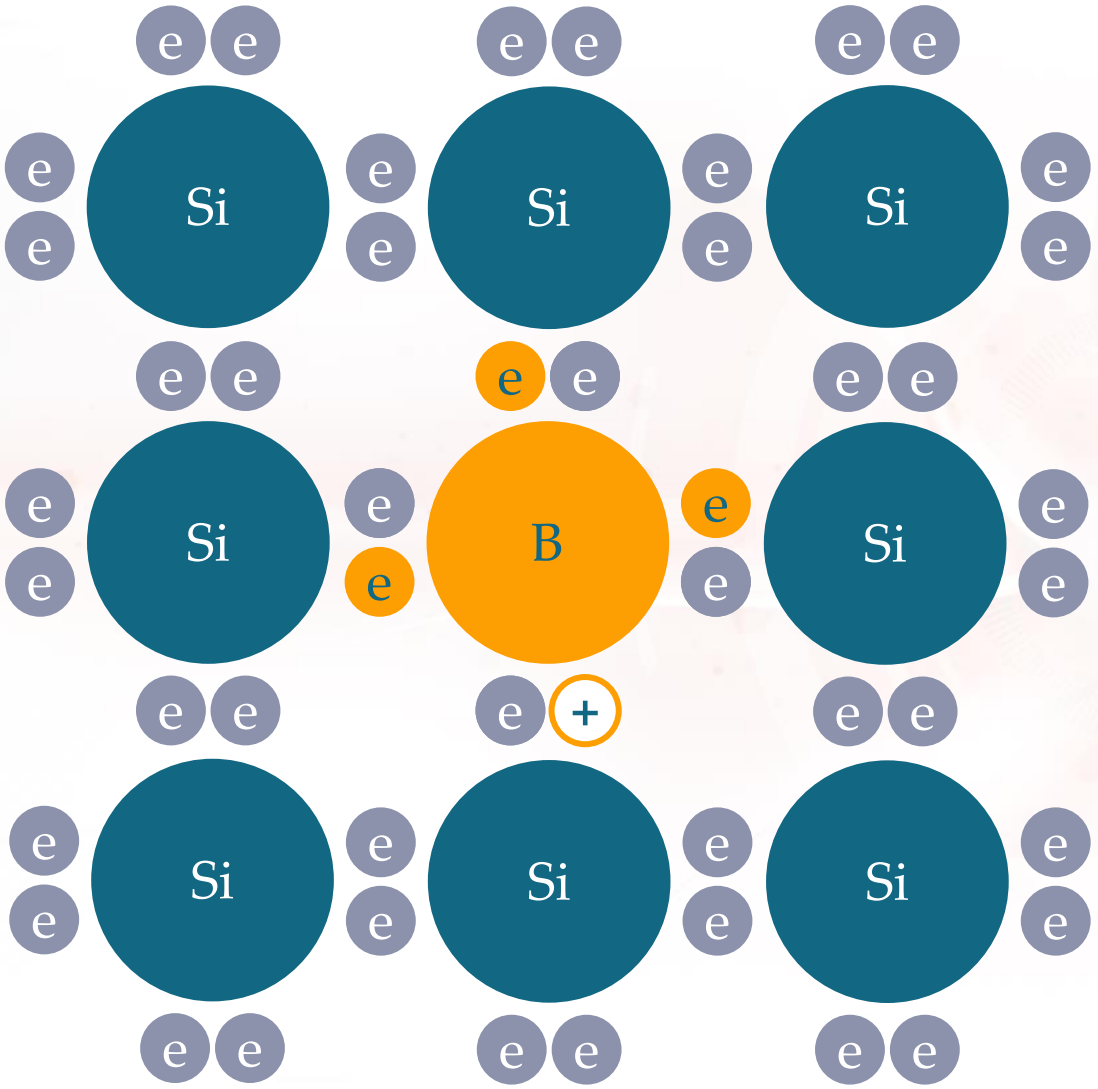
P-TYPE SEMICONDUCTOR



Doped with **trivalent** impurity

- Aluminum (Al)
- Boron (B)
- Indium (In)

P-TYPE SEMICONDUCTOR



Doped with **trivalent** impurity

- Aluminum (Al)
- Boron (B)
- Indium (In)

Majority Carriers	Minority Carriers
holes	electrons

EXERCISE

A doped semiconductor has **10 billion silicon** atoms and **15 million pentavalent** atoms. If the ambient temperature is 25º C, how many **free electrons** and **holes** are there inside the semiconductor?

1 excess electron per pentavalent atom

15 million free electrons

0 hole

EXERCISE

In reference to the previous example, if **5 million trivalent** atoms are added instead of pentavalent atoms, how many **holes** are there inside the semiconductor?

1 hole per trivalent atom

5 million holes

ENERGY LEVELS

Conductor

Conduction Band

Energy Gap = 0 eV

Valence Band

Insulator

Conduction Band

Energy Gap > 5 eV

Valence Band

Semiconductor

Conduction Band

Energy Gap = 1.1 eV(Si)

Energy Gap = **0.67 eV (Ge)**

Energy Gap = **1.41 eV (GaAs)**

Valence Band

LABORATORY